

04-09-2025

Reg. Adjustment:

$$\bar{Y}_{D=1} - \bar{x}_{D=1}^T \hat{\beta}_0 = \widehat{ATT}$$

$\xrightarrow{\quad} t(\bar{x}) \approx a \approx bD$

Propensity Score Reweighting:

$$\begin{aligned} E[Y(0) | D=1] &= E[E[Y(0) | X, D=0] | D=1] \\ &= E\left[ \frac{p(x)(1-p)}{p(1-p(x))} E[Y | X, D=0] | D=0 \right] \\ &= E\left[ \frac{p(x)(1-p)}{p(1-p(x))} Y | D=0 \right] \\ &\rightarrow \text{L.I.E in reverse} \end{aligned}$$

$$ATT = E[Y | D=1] - E[Y(0) | D=1]$$

$$= E\left[\frac{D}{p} Y\right] - E\left[\frac{p(x)(1-D)}{p(1-p(x))} Y\right]$$

$$= E\left[\underbrace{\left(\frac{D}{p} - \frac{p(x)(1-D)}{p(1-p(x))}\right)}_{\text{weights}} Y\right]$$

↳ weighted avg of Y's

$$\widehat{ATT} = \frac{1}{n} \sum_{i=1}^n \left( \frac{D_i}{\hat{p}} - \frac{\hat{p}(x_i)(1-D_i)}{\hat{p}(1-\hat{p}(x_i))} \right) Y_i$$

where  $\hat{p} = \bar{D}, \quad \frac{n_1}{n}$

$\hat{p}(x_i) \Rightarrow$  est. p-score in a first step

↳ Probit of D on X.

Do P-Score weighting for modelling assignment  
RA for modelling outcomes

## Augmented Inverse Propensity Score Weighting

You can show:

$$ATT = E \left[ \left( \frac{D}{P} - \frac{P(X)(1-D)}{P(1-P(X))} \right) (Y - E[Y|X, D=0]) \right] (*)$$

This estimator is doubly robust.

Consistent for ATT if at least 1/2 are correctly specified.

$$\begin{aligned} A &= E \left[ \frac{D}{P} (Y - E[Y|X, D=0]) - E \left[ \frac{P(X)(1-D)}{P(1-P(X))} (Y - E[Y|X, D=0]) \right] \right] \\ &= E[(Y - E[Y|X, D=0]) | D=1] - E \left[ \frac{P(X)(1-D)}{P(1-P(X))} (Y - E[Y|X, D=0]) | D=0 \right] \\ &= E[Y|X, D=1] - E[E[Y|X, D=0] | D=1] \\ &\quad - E \left[ \frac{P(X)(1-D)}{P(1-P(X))} \{ E[Y|X, D=0] - E[Y|X, D=0] \} | D=0 \right] \\ &\quad = 0 \end{aligned}$$

$p(x; \theta^*) :=$  model for prop score

$m(x; \beta_0^*) :=$  model for  $E[Y|X, D=0]$

→ in notes:  $\theta^*, \beta^*$  are pseudo-true values of the parameters

$$\begin{aligned} \hat{ATT} &= \frac{1}{n} \sum_{i=1}^n \left( \frac{D_i}{\hat{P}} - \frac{p(x_i; \hat{\theta})(1-D_i)}{\hat{p}(1-p(x_i; \hat{\theta}))} \right) (Y_i - m(x_i, \hat{\beta})) \\ \hat{ATT} &\xrightarrow{P} ATT^* \end{aligned}$$

s.t.

$$\begin{aligned} ATT^* &= E \left[ \frac{D}{P} - \frac{P(X, \theta^*)(1-D)}{P(1-P(X, \theta^*))} \right] (Y - m(X, \beta^*)) \\ &\quad \vdots \\ &= E[Y - m(X, \beta^*) | D=1] - E \left[ \frac{P(X, \theta^*)(1-D)}{P(1-P(X, \theta^*))} (Y - m(X, \beta^*)) | D=0 \right] \end{aligned}$$

Case 1: Outcome Reg Correct Spec

$$m(x; \beta^*) = E[Y|X, D=0] \\ \text{but } p(x; \theta^*) \neq p(x)$$

Becomes ATT-0 like before

Case 2: P-Score Model Correct Spec:

$$\begin{aligned} p(x; \theta^*) &= p(x) \\ \text{but } m(x; \beta^*) &\neq E[Y|X, D=0] \\ &= \dots - E\left[\left(\frac{p(x)(1-p)}{p(1-p(x))}\right)(Y - m(x; \beta^*)) \mid D=0\right] \\ &= \dots - E\left[\frac{p(x)(1-p)}{p(1-p(x))} (E[Y|X, D=0] - m(x; \beta^*)) \mid D=0\right] \\ &= \dots - E[E[Y|X, D=0] - m(x; \beta^*) \mid D=1] \\ &= E[Y|X, D=1] - E[E[Y|X, D=0] \mid D=1] \\ &\hookrightarrow \text{Removes } m(x; \beta^*) \text{ via subtraction} \end{aligned}$$